

* Unit - 1 * (Fundamental)

* Introduction to mech. Engg.:-

mechanical engg. is one of the core branches of engg. that deals with the design, analysis, manufacturing and maintenance of mechanical systems. It combines principles from physics & materials science to develop machinery & mechanical systems for various application in industries, automobiles, aerospace, energy & more.

Concept of Thermal engg.:-

Thermal engg focuses on the study of heat energy & its transfer, conversion & management in mechanical system.

it includes -

- Thermodynamics
- Heat transfer mechanism

- ↳ Conduction
- ↳ Convection
- ↳ Radiation

- Thermal system - Engine, Refrigerators, power plant, IC engine.

Conservation of energy - energy can't be created or destroyed. only transformed from one to another.

System - A quantity of matter or a region in space chosen for study.

Surrounding - The mass or region outside the system.

Boundary - Real / imaginary

That separate the system & surrounding



* Types of system.

Isolated system

Neither mass nor energy can cross the boundary.

Ex well-insulated, closed thermos bottle

Closed system

Only energy can cross the boundary

Ex A tightly capped cup of coffee.

Open system

Only both mass & energy can cross boundary.

Ex Air compressor

Thermodynamic properties -

- Intensive - Are independent of amount of mass.

Ex Temp, Pressure, density

- Extensive - varies directly with the mass

Ex - mass, volume, energy, enthalpy

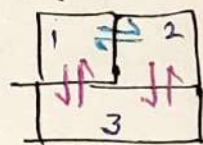
$$\text{Intensive} = \frac{\text{Extensive}}{m}$$

(Specific Properties)

Thermodynamic Law

(1) Zeroth law of Thermodynamic -

" If two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other.



(2) First law of Thermo -

(Energy conservation law)

In thermodynamic system,

net heat supplied to the system from surrounding is equal to net work done by system on surrounding.

$$dW = pdV \quad \longleftrightarrow \quad dU = dQ - dW$$

Internal energy
enthalpy
work done

Second law of thermo-

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According to clausius - It is impossible to construct a device which operates on a cycle to which can convert whole work done into heat. Some losses of heat occurs.

in other way, Efficiency cannot be 100%.

* Thermodynamic Process

- Reversible process - System & surrounding can be returned to their original state.

$\Delta \text{entropy} = 0$, cyclic process. (equilibrium state)

- Irreversible Process - Non-equilibrium state. Initial point & final point are different.

- Cyclic Process - Initial & final point are same. total internal energy = 0

$$\boxed{\Delta U = 0}$$

- Isothermic Process -

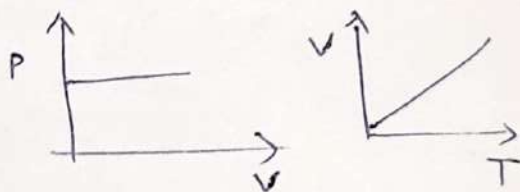
Pressure remains constant

$$P_1 = P_2$$

$$\Delta P = 0$$

$$\text{work done (W)} = \Delta V \cdot P$$

$$PV = nRT \Rightarrow (V \propto T)$$



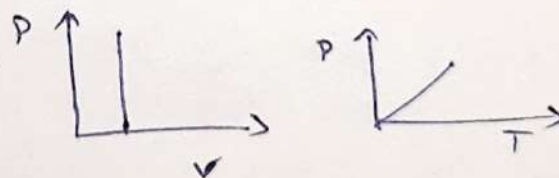
- Isochoric Process -

Volume remains constant.

$$\boxed{W = 0}$$

$$\Delta V = 0$$

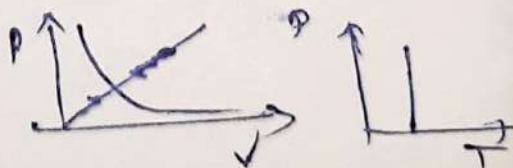
$$PV = nRT \Rightarrow P \propto T$$



- Isothermal Process -

Temp. remains constant $\Delta T = 0$

$$\boxed{W = nRT \ln \left(\frac{T_2}{T_1} \right)}$$



• Adiabatic Process -

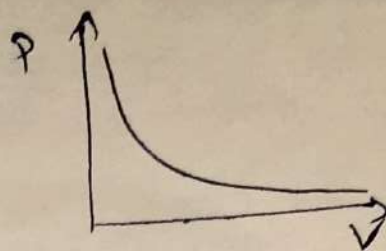
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No Heat exchange

$$\boxed{\Delta Q = 0}$$

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$\gamma = \frac{C_p}{C_v} \quad \& \quad \boxed{PV^\gamma = \text{Constant}}$$



* Heat transfer → Transfer of heat between bodies or within a body due to temp difference.

modes

Conduction Transfer of heat through a solid from high to low temp. (without any movement)
 $(Q = -KA \frac{dT}{dx})$

Convection Transfer of heat in a fluid due to combined effect of fluid motion & conduction.
 $(Q = hA(T_s - T_\infty))$

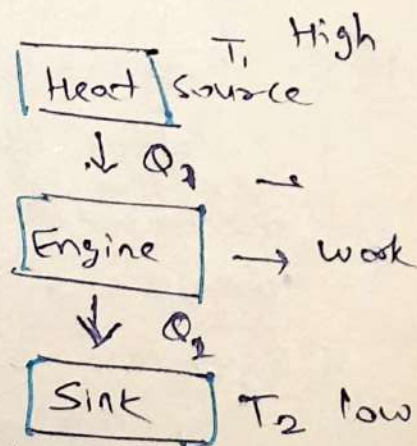
Radiation - Transfer of heat through electromagnetic waves without any medium.
 $\boxed{Q = \epsilon \sigma A T^4}$

* Heat engine - Converts heat energy → Mechanical energy.

↳ hot source
 ↳ cold sink.

working

- Absorbs Q_1 of hot source
- convert into work (W)
- Remaining Q_2 is rejected to sink.
- working fluid returns into initial state & cycle →



Efficiency:- ratio of work output to heat input 39

$$\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

for cannot heat engine.

$$\eta = 1 - \frac{T_2}{T_1} \quad \begin{matrix} (T_2 < T_1) \\ \downarrow \quad \uparrow \\ \text{Sink} \quad \text{Heat} \end{matrix}$$

Note Automobile $\Rightarrow$ IC engines.

Power plants $\Rightarrow$ Steam turbines.

* Mechanical machine design:- mechanical machine design involves the creation of mechanical components & systems that meet specific functional requirement under defined constraints.

- $\rightarrow$ Design principle - stress-strain analysis, factor of safety.
- $\rightarrow$ Machine element - gear, bearing, shaft, spring.

* Industrial engineering - It aims to optimize complex processes, systems or organizations. It involves

- Operation research.
- Production planning
- Quality control & supply chain management.
- Focuses on efficiency, productivity, cost reduction.

Note Properties of steam -

Vapor form of water, widely used in mechanical & thermal engg.

pressure, temp, enthalpy, entropy, internal energy.

Dryness fraction.

(0 to 1)

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* Steam boiler:- It is a closed vessel in which water is heated to produce steam using the heat energy from combustion of fuel. Generated steam is used in power generation, industrial heating & mechanical work.

main components

- Furnace - Provide heat by fuel combustion.
- Safety valve - avoid over pressure.
- Steam stop valve - stop & regulate steam flow.
- Vent valve - To vent air from steam drum during starting.
- pressure gauge → To indicate steam pressure inside.
- Boiler Drum - Stores water & steam.

Working [Same for all type of boiler]

(1) Fuel combustion in Furnace -

→ Fuel is burned in the Furnace.

This combustion release a large amount of heat energy.

(2) Heat transfer to water -

→ The heat from the burning fuel is transfer to water stored in the boiler's drum or shell.

It occurs due to conduction / convection.

(3) Water converts to steam -

→ As water absorbs heat, its temp ↑ & eventually reaches at boiling point.

→ water evaporates into steam.

(4) Steam collection & Delivery

→ The steam rises & collects in the steam space above water level.

→ It is ~~drawn~~ drawn out via steam stop valve.

(5) Fuel gases exit through chimney.

* Classification of steam boiler

- Based on Axis
Horizontal & Vertical boiler
- Based on Flow of water & Hot gases
Fire tube & water tube boiler
- Based on pressure
Low pressure & High pressure

<u>Boiler type</u>	<u>Feature</u>
• Cochran boiler	- Verticle, <u>fire tube</u> , Compact shaped.
• Lancashire boiler	- Horizontal, <u>fire tube</u> , large shaped.
• Babcock & wilcox	- Water tube, high pressure, efficient Horizontal.
• Locomotive Boiler	- Horizontal, mobile, <u>fire tube</u> .

Horizontal Boiler - Horizontal shell & tubes.

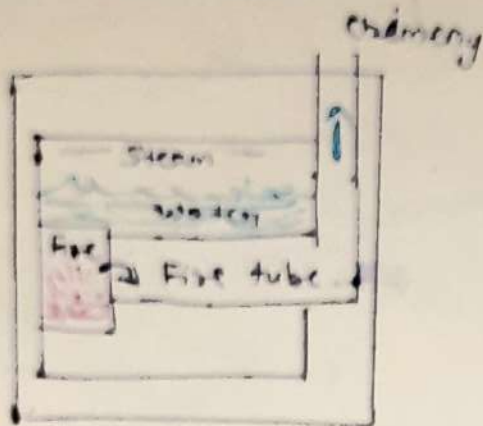
High efficiency, steam output high, longer size, easy maintenance

Vertical Boiler → Verticle shell & tubes.

low efficiency, steam output low, compact size, Harder maintenance.

* Fire tube and Water tube boiler

Fire tube boiler



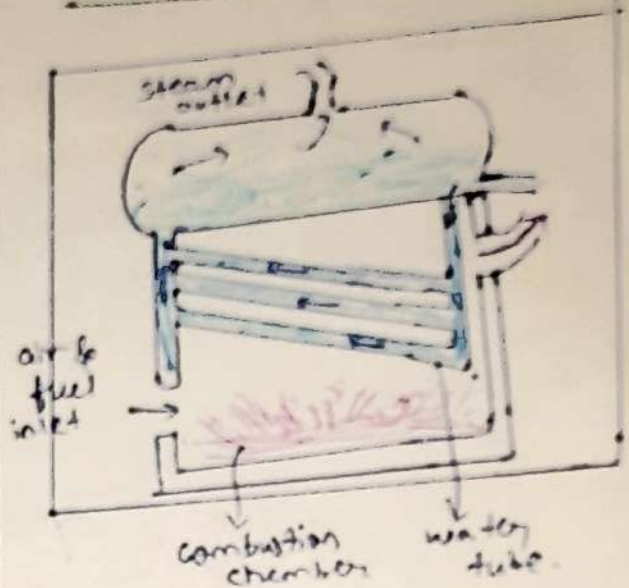
Ex

- i) It is compact construction
- ii) Capacity range = 12000 kg/hr
- iii) Obtained 18 kg/cm² pressure
- iv) transportation is easier.
- v) It is cheap in cost
- vi) Time taken is large & less efficiency.
- vii) Fuel gases (hot) in pipes water outside
- viii) Rate of steam production is lower
- ix) Water treatment is not so necessary.

~~used for~~ for

Ex Cochran, locomotive boiler, Lancashire boiler

Water tube boiler



- i) It is not compact.
- ii) Capacity range = 4500 to 12000 kg/hr
- iii) Very high pressure in order 140 kg/cm² obtained.
- iv) Transportation is difficult.
- v) Expensive in cost.
- vi) large heating surface & high efficiency
- vii) water in pipes. fire out
- viii) Rate of steam production is higher.
- ix) Water treatment is necessary.

* Used for high steam demand & pressure requirement.

Ex Babcock, Wilcox, boiler.

Selection of a boiler -

- Working pressure & quality of steam
- Boiler generation rate
- Floor area available
- Accessibility for repair & inspection.
- The fuel & water available.
- Operating & maintenance cost.

* Steam Turbine :- A mechanical device that converts thermal energy of steam into mechanical work (rotational energy).

Component -

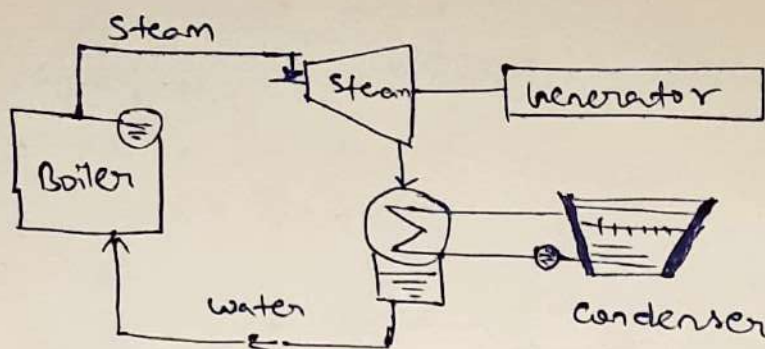
- Nozzles - Converts steam pressure into high velocity jets.
- Rotor (Shaft) - Hold the rotating blade & transmit torque.
- Blades - Steam strike them to rotate.
- Casing - Encloses the turbine & maintain pressure.
- Bearing - Support the rotor & reduce friction.
- Governor - Control speed of Turbine.
- Condenser - Converts exhaust steam back to water.

* Working - High pressure steam from the boiler enters the turbine.

- Steam is directed onto blades & nozzles arranged on a rotor.
- The force of steam pushes the blades, causing the rotor to spin.
- As the steam expands, it loses pressure & temp.
- The shaft rotation is used to drive a generator pump or mechanical equipment.

Type Impulse Turbine - steam expansion in nozzle only, hits moving blades with high velocity.

Reaction Turbine - Steam expands in both fixed & moving blades.



Advantage - High efficiency for large power generation
 - Smooth & continuous rotary motion.
 - Suitable for high pressure & temp² condition
 - Low maintenance & long life.

* Power Plant Power plant is an industrial unit where electric power is generated from natural or artificial energy source. Then transmitted to homes industries & commercial facilities.

Types - Nuclear Power Plant
 Thermal Power Plant
 Diesel Power Plant
 Steam Power Plant.

(1) Nuclear Power Plant - Heat energy is generated by splitting atoms in a reactor, which then produce steam & generate electricity by steam turbine.

Components

Nuclear reactor - Site of the fission reaction, generates heat.

Fuel rods - Contain enriched uranium or plutonium, U^{235} or Pu^{239}

Moderator - Slow down neutrons Commonly Heavy water (D_2O) or graphite.

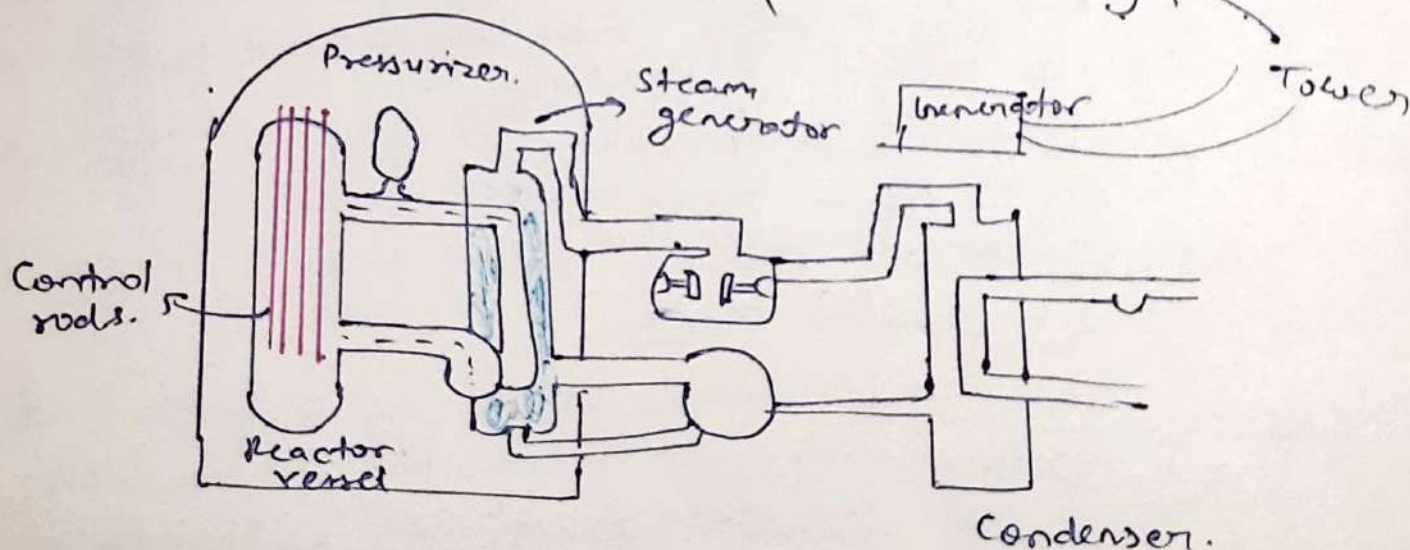
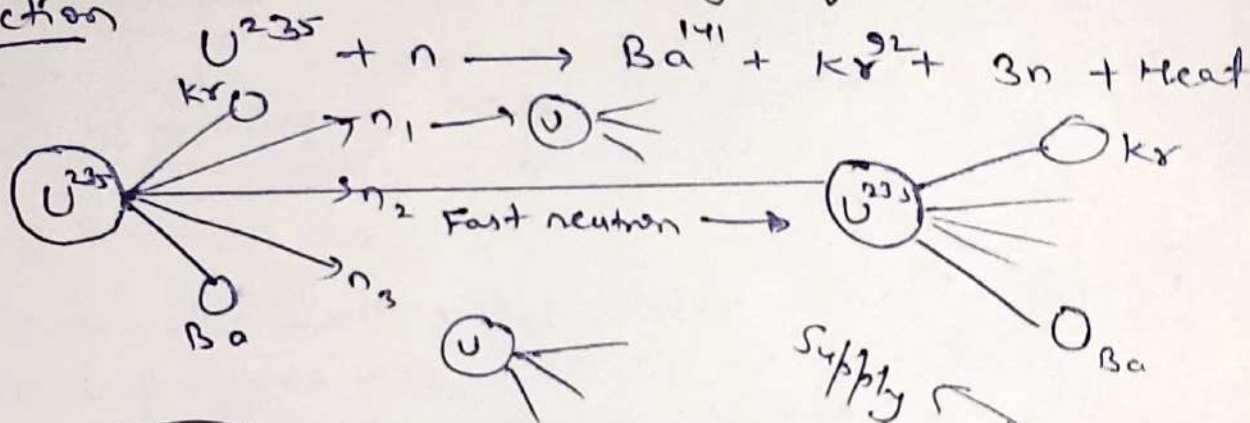
Control rods - Absorbs excess neutrons to regulate the reaction. (Boron / Cd)

Coolant - Transfer heat from reactor to steam generator.

Steam generator, condenser.

Containment Building - Sealed structure to prevent leakage of radiation.

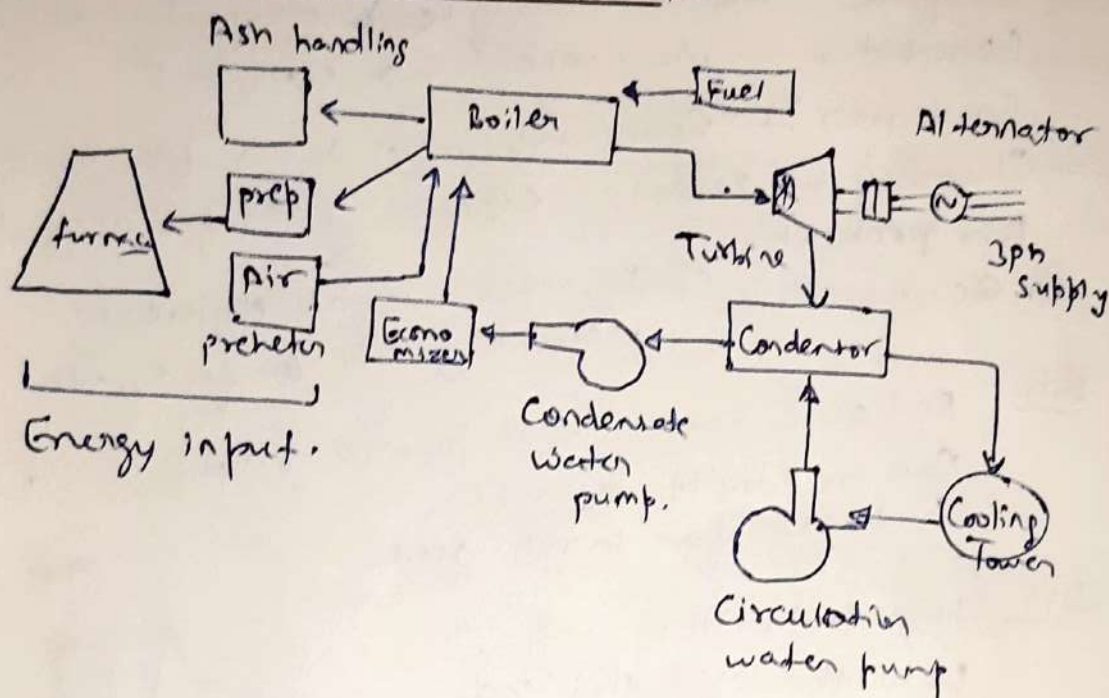
Reaction



- Advantage emit low amount of CO_2 ,
 → emission of green house gases less.
 → Readily available.
 → generate high amount of energy.

- Disadvantage → Radioactive waste
 → High Risk.
 → It can be misused as terrorist attack or nuclear weapons.

(Q) Thermal (Steam) Power Plant:-



Power generating system that converts chemical energy of fuel into thermal energy → mechanical energy. → electrical energy.

Working .

- Fuel combustion → coal is burned in boiler to produce heat.
- Steam generator → heat converts water into high pressure steam.
- Steam expansion → Steam is directed into blade of a turbine, causing it to spin.
- power generation → turbine shaft + generator ⇒ Power

- Condensation exhaust steam is passed into a condenser, where it is cooled & condensed back into water.
- Water circulation - Feed pump sends the condensed water back to the boiler.

components

- Boiler - Converts water into steam
- Superheater - Increase temp of steam
- Turbine - thermal energy $\longleftrightarrow$ mechanical energy
- Generator - mechanical $\rightarrow$ electrical
- Condenser - Condenses ~~water~~ steam into water.
- Chimney - Release exhaust gases from combustor
- Air preheater & Economiser $\rightarrow$ Increase overall efficiency by preheating air & feedwater

Adv

- Reliable base-load power supply
- Can be built anywhere.
- Suitable for large scale

Dis

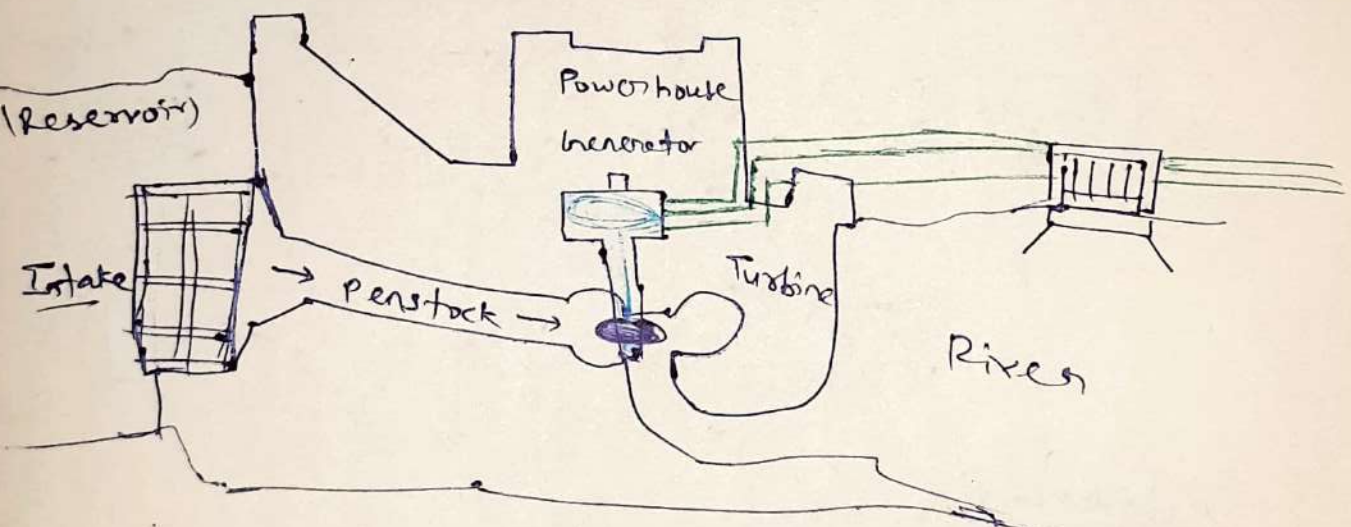
- High pollution
- Low thermal efficiency.
- High water consumption.

Hydroelectric Power Plant :-

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It is a renewable energy facility that generates electricity by harnessing the energy of falling or flowing water to spin a turbine connected to a generator working

- Water storage - water stored at height in a dam reservoir.
- Water flow - Water is released through a penstock
- Turbine Rotation - The force of water turns the turbine blades at the base.
- Electricity generation - The rotating turbine shafts spins a generator, producing electricity
- Water Discharge - After passing through the turbine water is discharged back into the river



Dam $\Rightarrow$ store water at a height to create potential energy.

Penstock $\Rightarrow$ A pressure conduit that carries water from the reservoir to the turbine

Surge tank $\Rightarrow$ Controls pressure variation in penstock

Diesel power plant

A diesel power plant is a power generating station in which a diesel engine is used to drive an alternator for generating electricity.

Working

Air intake - Air is drawn into the engine cylinder.

Compression - The piston compresses the air, raising its temperature.

Fuel injection - Diesel is injected and auto ignites due to high temp.

Power stroke - Explosion forces piston downward, rotation the crankshaft.

Exhaust - Burnt gases are expelled through the valve.

Power generation - The rotating crankshaft turns the alternator to generate electricity.

